Athenaサンプルクエリ

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テーブル作成

※クエリ実行前に以下の変数を書き換えること。

${テーブル名}	任意の名前
${環境名}	test/prod/dr-prod

バウンス用

- 対象イベントタイプ
  - Bounce
  - DeliveryDelay

```
CREATE EXTERNAL TABLE ${テーブル名} (
    eventType string,
    mail struct <
```

```
        timestamp: string,
        source: string,
        sourcearn: string,
        sendingaccountid: string,
        messageid: string,
        destination: string,
        headerstruncated: boolean,
        headers: array < struct <
            name: string,
            value: string
            >
    >,
        commonheaders: struct <
            `from`: array < string >,
             to: array < string >,
             messageid: string,
             subject: string
        >,
        tags: struct <
             ses_source_tls_version: string,
             ses_operation: string,
             ses_configurationset: string,
             ses_source_ip: string,
             ses_outgoing_ip: string,
             ses_from_domain: string,
             ses_caller_identity: string
            >
    >,
    bounce struct <
        bouncedrecipients: array < struct <
            action: string,
            diagnosticcode: string,
            emailaddress: string,
            status: string
            >
        >,
        bouncesubtype: string,
        bouncetype: string,
        feedbackid: string,
        remotemtaip: string,
        reportingmta: string,
        `timestamp`: string
    >,
    deliverydelay struct <
        delaytype: string,
        expirationtime: string,
        delayedrecipients: array < struct <
            emailaddress: string,
            status: string,
            diagnosticcode: string >
        >
    >
)
PARTITIONED BY (
`partition_date` string
)
ROW FORMAT SERDE 'org.openx.data.jsonserde.JsonSerDe'
WITH SERDEPROPERTIES (
    "mapping.ses_caller_identity" = "ses:caller-identity",
    "mapping.ses_configurationset" = "ses:configuration-set",
    "mapping.ses_from_domain" = "ses:from-domain",
```

```
    "mapping.ses_operation" = "ses:operation",
    "mapping.ses_outgoing_ip" = "ses:outgoing-ip",
    "mapping.ses_source_ip" = "ses:source-ip",
    "mapping.ses_source_tls_version" = "ses:source-tls-version",
    "mapping.ses_sender_identity" = "ses:sender-identity"
)
LOCATION 's3://managed-smail-${環境名}-s3-bucket-bounce-event/'
TBLPROPERTIES (
  'projection.enabled'='true',
  'projection.partition_date.type' = 'date',
  'projection.partition_date.range' = '2020/10/01,NOW',
  'projection.partition_date.format' = 'yyyy/MM/dd',
  'projection.partition_date.interval' = '1',
  'projection.partition_date.interval.unit' = 'DAYS',
  'storage.location.template' = 's3://managed-smail-${環境名}-s3-bucket-bounce-event/${partition
```

その他イベント用

- 対象イベントタイプ
  - Delivery
  - Complaint
  - Reject
  - Send

```
CREATE EXTERNAL TABLE ${テーブル名} (
    eventType string,
    mail struct <
        timestamp: string,
        source: string,
        sourcearn: string,
        sendingaccountid: string,
        messageid: string,
        destination: string,
        headerstruncated: boolean,
        headers: array < struct <
            name: string,
            value: string
            >
    >,
        commonheaders: struct <
            `from`: array < string >,
             to: array < string >,
             messageid: string,
             subject: string
        >,
        tags: struct <
             ses_source_tls_version: string,
             ses_operation: string,
             ses_configurationset: string,
             ses_source_ip: string,
             ses_outgoing_ip: string,
             ses_from_domain: string,
             ses_caller_identity: string
```

```
            >
    >,
    delivery struct <
        reportingmta: string,
        smtpresponse: string,
        remotemtaip: string,
        processingtimemillis: int,
        `recipients`: array < string >
    >,
    complaint struct <
        feedbackid: string,
        useragent: string,
        complaintfeedbacktype: string,
        arrivaldate: string,
        complainedrecipients: array < struct <
            emailaddress: string >
        >
    >,
    reject struct <
        reason: string
    >
)
PARTITIONED BY (
`partition_date` string
)
ROW FORMAT SERDE 'org.openx.data.jsonserde.JsonSerDe'
WITH SERDEPROPERTIES (
    "mapping.ses_caller_identity" = "ses:caller-identity",
    "mapping.ses_configurationset" = "ses:configuration-set",
    "mapping.ses_from_domain" = "ses:from-domain",
    "mapping.ses_operation" = "ses:operation",
    "mapping.ses_outgoing_ip" = "ses:outgoing-ip",
    "mapping.ses_source_ip" = "ses:source-ip",
    "mapping.ses_source_tls_version" = "ses:source-tls-version",
    "mapping.ses_sender_identity" = "ses:sender-identity"
)
LOCATION 's3://managed-smail-${環境名}-s3-bucket-mail-event/'
TBLPROPERTIES (
  'projection.enabled'='true',
  'projection.partition_date.type' = 'date',
  'projection.partition_date.range' = '2020/10/01,NOW',
  'projection.partition_date.format' = 'yyyy/MM/dd',
  'projection.partition_date.interval' = '1',
  'projection.partition_date.interval.unit' = 'DAYS',
  'storage.location.template' = 's3://managed-smail-${環境名}-s3-bucket-mail-event/'
)
```

ビュー作成

※クエリ実行前に以下の変数を書き換えること。

${テーブル名}	作成したテーブル名

ハードバウンス用

```
CREATE OR REPLACE VIEW bounce AS
SELECT
partition_date,
eventType,
bounce.feedbackid,
bounce.bouncetype,
bounce.bouncesubtype,
  TRY(bounce.bouncedrecipients[1].emailaddress) AS bounce_recipient_email,
  TRY(bounce.bouncedrecipients[1].action) AS bounce_recipient_action,
  TRY(bounce.bouncedrecipients[1].status) AS bounce_recipient_status,
  TRY(bounce.bouncedrecipients[1].diagnosticCode) AS bounce_recipient_diagnosticcode,
bounce.timestamp,
bounce.remoteMtaIp,
bounce.reportingMTA,
mail.source,
mail.sourceArn,
mail.sendingAccountId,
mail.messageId,
mail.destination,
mail.headersTruncated,
  TRY(filter(mail.headers, h -> h.name = 'From')[1].value) AS from_address,
  TRY(filter(mail.headers, h -> h.name = 'To')[1].value) AS to_address,
  TRY(filter(mail.headers, h -> h.name = 'Subject')[1].value) AS subject,
  TRY(filter(mail.headers, h -> h.name = 'MIME-Version')[1].value) AS mine_version,
  TRY(filter(mail.headers, h -> h.name = 'Content-Type')[1].value) AS content_type,
mail.tags.ses_source_tls_version,
mail.tags.ses_operation,
mail.tags.ses_source_ip,
mail.tags.ses_from_domain,
mail.tags.ses_caller_identity
FROM
${テーブル名}
WHERE (eventtype = 'Bounce')
```

ソフトバウンス（配信遅延）用

```
CREATE OR REPLACE VIEW delivery_delay AS
SELECT
partition_date,
eventType,
mail.timestamp,
mail.source,
mail.sourceArn,
mail.sendingAccountId,
mail.messageId,
mail.destination,
mail.headersTruncated,
mail.tags.ses_source_tls_version,
mail.tags.ses_operation,
mail.tags.ses_source_ip,
mail.tags.ses_from_domain,
mail.tags.ses_caller_identity,
deliverydelay.delaytype,
deliverydelay.expirationtime,
  TRY(deliverydelay.delayedrecipients[1].emailaddress) AS deliverydelay_emailaddress,
  TRY(deliverydelay.delayedrecipients[1].status) AS deliverydelay_status,
  TRY(deliverydelay.delayedrecipients[1].diagnosticcode) AS deliverydelay_diagnosticcode
```

```
FROM
${テーブル名}
WHERE (eventtype = 'DeliveryDelay')
```

配信成功用

```
CREATE OR REPLACE VIEW delivery AS
SELECT
partition_date,
eventType,
mail.timestamp,
mail.source,
mail.sourceArn,
mail.sendingAccountId,
mail.messageId,
mail.destination,
mail.headersTruncated,
  TRY(filter(mail.headers, h -> h.name = 'From')[1].value) AS from_address,
  TRY(filter(mail.headers, h -> h.name = 'To')[1].value) AS to_address,
  TRY(filter(mail.headers, h -> h.name = 'Subject')[1].value) AS subject,
  TRY(filter(mail.headers, h -> h.name = 'MIME-Version')[1].value) AS mine_version,
  TRY(filter(mail.headers, h -> h.name = 'Content-Type')[1].value) AS content_type,
mail.tags.ses_source_tls_version,
mail.tags.ses_operation,
mail.tags.ses_source_ip,
mail.tags.ses_from_domain,
mail.tags.ses_caller_identity,
delivery.processingtimemillis,
delivery.recipients,
delivery.smtpresponse,
delivery.reportingmta
FROM
${テーブル名}
WHERE (eventtype = 'Delivery')
```

苦情用

```
CREATE OR REPLACE VIEW complaint AS
SELECT
partition_date,
eventType,
mail.timestamp,
mail.source,
mail.sourceArn,
mail.sendingAccountId,
mail.messageId,
mail.destination,
mail.headersTruncated,
  TRY(filter(mail.headers, h -> h.name = 'From')[1].value) AS from_address,
  TRY(filter(mail.headers, h -> h.name = 'To')[1].value) AS to_address,
  TRY(filter(mail.headers, h -> h.name = 'Subject')[1].value) AS subject,
  TRY(filter(mail.headers, h -> h.name = 'MIME-Version')[1].value) AS mine_version,
  TRY(filter(mail.headers, h -> h.name = 'Content-Type')[1].value) AS content_type,
mail.tags.ses_source_tls_version,
```

```
mail.tags.ses_operation,
mail.tags.ses_source_ip,
mail.tags.ses_from_domain,
mail.tags.ses_caller_identity,
complaint.feedbackid,
complaint.useragent,
complaint.complaintfeedbacktype,
complaint.arrivaldate,
  TRY(complaint.complainedrecipients[1].emailaddress) AS complainedrecipients_emailaddress
FROM
${テーブル名}
WHERE (eventtype = 'Complaint')
```

拒否用

```
CREATE OR REPLACE VIEW reject AS
SELECT
partition_date,
eventType,
mail.timestamp,
mail.source,
mail.sourceArn,
mail.sendingAccountId,
mail.messageId,
mail.destination,
mail.headersTruncated,
  TRY(filter(mail.headers, h -> h.name = 'From')[1].value) AS from_address,
  TRY(filter(mail.headers, h -> h.name = 'To')[1].value) AS to_address,
  TRY(filter(mail.headers, h -> h.name = 'Subject')[1].value) AS subject,
  TRY(filter(mail.headers, h -> h.name = 'MIME-Version')[1].value) AS mine_version,
  TRY(filter(mail.headers, h -> h.name = 'Content-Type')[1].value) AS content_type,
mail.tags.ses_source_tls_version,
mail.tags.ses_operation,
mail.tags.ses_source_ip,
mail.tags.ses_from_domain,
mail.tags.ses_caller_identity,
reject.reason
FROM
${テーブル名}
WHERE (eventtype = 'Reject')
```

送信用

```
CREATE OR REPLACE VIEW send AS
SELECT
partition_date,
eventType,
mail.timestamp,
mail.source,
mail.sourceArn,
mail.sendingAccountId,
mail.messageId,
mail.destination,
mail.headersTruncated,
```

```
  TRY(filter(mail.headers, h -> h.name = 'From')[1].value) AS from_address,
  TRY(filter(mail.headers, h -> h.name = 'To')[1].value) AS to_address,
  TRY(filter(mail.headers, h -> h.name = 'Subject')[1].value) AS subject,
  TRY(filter(mail.headers, h -> h.name = 'MIME-Version')[1].value) AS mine_version,
  TRY(filter(mail.headers, h -> h.name = 'Content-Type')[1].value) AS content_type,
mail.tags.ses_source_tls_version,
mail.tags.ses_operation,
mail.tags.ses_source_ip,
mail.tags.ses_from_domain,
mail.tags.ses_caller_identity
FROM
${テーブル名}
WHERE (eventtype = 'Send')
```

結果抽出

・以下クエリの${ビュー名}を下記表のいずれかに置き換える。

イベントタイプ	${ビュー名}
ハードバウンス	bounce
ソフトバウンス	delivery_delay
配信成功	delivery
苦情	complaint
拒否	reject
送信	send

年月日/期間のみを指定

・特定の日（2025/09/01）のログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE partition_date = '2025/09/01';
```

・特定の期間（2025/09/01～2025/09/30）のログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE
  partition_date BETWEEN '2025/09/01' AND '2025/09/30';
```

※あまり長い期間を選択してしまうと、大量のS3オブジェクトに対してGetイベントが発生するため、S3のオブジェクト取得費用やCloudTrailのデータイベント費用等が増加する点に注意。

送信先メールアドレスを指定

・年月日と送信先メールアドレスを指定してログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE
  partition_date = '2025/09/01' AND destination LIKE '%xxx@xxx.com%';
```

・期間と送信先メールアドレスを指定してログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE
  partition_date BETWEEN '2025/09/01' AND '2025/09/30' AND destination LIKE '%xxx@xxx.com%';
```

送信元IPを指定

※delivery_delayイベントはログに「ses_source_ip」が含まれない可能性があるため、delivery_delayイベントが発生したメールの送信元IPを調査する場合は、
delivery_delayイベントが発生したメールの「messageId」を控えておき、sendイベントに対して「messageId」を指定する形でクエリを実行する。

・年月日と送信元IPを指定してログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE
  partition_date = '2025/09/01' AND ses_source_ip LIKE '%192.168.3.9%';
```

・期間と送信元IPを指定してログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE
  partition_date BETWEEN '2025/09/01' AND '2025/09/30' AND ses_source_ip LIKE '%192.168.3.9%';
```

件名を指定

※delivery_delayイベントはログに「header」および「subject」が含まれない可能性があるため、delivery_delayイベントが発生したメールの件名を調査する場合は、
delivery_delayイベントが発生したメールの「messageId」を控えておき、sendイベントに対して「messageId」を指定する形でクエリを実行する。

・年月日と件名を指定してログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE
  partition_date = '2025/09/01' AND subject = 'test-mail';
```

・期間と件名を指定してログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE
  partition_date BETWEEN '2025/09/01' AND '2025/09/30'  AND subject = 'test-mail';
```

messageIdを指定

・年月日とmessageIdを指定してログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE
  partition_date = '2025/09/01' AND messageid = 'xxxxxxx';
```

・期間とmessageIdを指定してログを抽出する場合

```
SELECT
  *
FROM
  ${ビュー名}
WHERE
  partition_date BETWEEN '2025/09/01' AND '2025/09/30'  AND messageid = 'xxxxxxx';
```

テーブル削除

```
DROP TABLE ${テーブル名};
```

ビュー削除

・以下クエリの${ビュー名}を下記表のいずれかに置き換える。

イベントタイプ	${ビュー名}
ハードバウンス	bounce
ソフトバウンス	delivery_delay
配信成功	delivery
苦情	complaint
拒否	reject
送信	send

```
DROP VIEW ${ビュー名};
```